

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 8: Measuring Growth: Seedling Height, Stem Width and Growth Curves of Maize and Beans

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	D	A	C	B	A	A

USE AFTER: PREDICT PHASE — Predict: sketching maize and bean height over 21 days (Think-Pair-Share)

Q1. Before measuring anything, a learner predicts that a maize seedling will grow the same number of centimetres every day for 21 days. Which statement about real seedling growth best challenges this prediction?

- A. Growth happens only in the first three days
- B. Growth keeps getting faster every day until the plant dies
- C. Growth is slow at first, then rapid, then slows down and levels off**
- D. Growth is steady, so the graph is a straight line

Correct answer: C. Real growth curves are S-shaped, with a lag phase, a rapid phase and a stationary phase, so the growth rate changes over time. A straight line wrongly assumes a constant daily increase.

USE AFTER: OBSERVE PHASE — Observe: measuring stem width with thread and ruler

Q2. A group wraps thread once around a bean stem. The thread measures 12 mm against a ruler. Using the method of dividing the circumference by 3, what is the approximate stem width (diameter)?

- A. 12 mm
- B. 9 mm
- C. 36 mm
- D. 4 mm**

Correct answer: D. Diameter is about circumference divided by 3, so $12 \div 3 = 4$ mm. 12 mm is the circumference itself, not the width.

USE AFTER: OBSERVE PHASE — Observe: measuring 3 maize and 3 bean seedlings and calculating mean values

Q3. A group measures three maize seedlings from the class nursery. Their heights are 30 cm, 32 cm and 40 cm. What is the mean height?

- A. 34 cm**
- B. 32 cm
- C. 102 cm
- D. 40 cm

Correct answer: A. Mean = $(30 + 32 + 40) \div 3 = 102 \div 3 = 34$ cm. Using several seedlings and finding the mean gives a more reliable value than one seedling.

USE AFTER: EXPLAIN PHASE — Explain: writing a CER statement using numbers from the table and graph

Q4. Mean height data: maize was 8 cm on Day 6 and 48 cm on Day 15. Beans were 3 cm on Day 6 and 17 cm on Day 15. Between Day 6 and Day 15, how many more centimetres did the maize grow than the beans?

- A. 40 cm
- B. 14 cm
- C. 26 cm**
- D. 31 cm

Correct answer: C. Maize grew $48 - 8 = 40$ cm. Beans grew $17 - 3 = 14$ cm. The difference is $40 - 14 = 26$ cm, which is evidence that maize grew faster in this period.

USE AFTER: EXPLAIN PHASE — Explain: interpreting lag, rapid and stationary phases on the growth curves

Q5. Mean maize height increased from 18 cm on Day 9 to 32 cm on Day 12. It then went from 58 cm on Day 18 to only 62 cm on Day 21. What does the flatter part of the curve between Day 18 and Day 21 show?

- A. The seed has not yet absorbed water
- B. The stationary phase, where growth slows as the plant matures or resources limit it**
- C. The rapid phase, where growth is fastest
- D. The plant has died

Correct answer: B. Growth of only 4 cm in 3 days, compared with 14 cm earlier, shows the stationary phase. The plant is still alive, but its growth has slowed.

USE AFTER: DQB PHASE — DQB: adding evidence and the new question 'Do we have hormones in plants?'

Q6. Maize and bean seedlings were planted on the same day after the same rain. The two growth curves show that growth is not constant, and that the two crops grow at different rates. Why does this lead to the question 'Do we have hormones in plants?'

- A. Because growth speeds up and slows down at different times, suggesting something inside the plant may control its timing and speed**
- B. Because seeds that failed to germinate have no growth curve
- C. Because the curves are straight lines
- D. Because plants make insulin like animals do

Correct answer: A. Changing growth rates under the same conditions suggest an internal control. This is what makes learners ask whether plants have hormones as animals do.

USE AFTER: MODEL BUILDING PHASE — Model: revising the Day 1 drawing with food store arrows, growth curve sketches and a Gallery Walk

Q7. In a revised model of early growth, arrows are drawn showing food stores feeding the young seedling. Which pair correctly shows where maize and bean seedlings get these early food stores?

- A. Maize from the endosperm, beans from the cotyledons**
- B. Both from hormones in the soil
- C. Both only from photosynthesis in the first days
- D. Maize from the cotyledons, beans from the endosperm

Correct answer: A. In maize, a monocot, the food is stored mainly in the endosperm. In beans, a dicot, it is stored in the cotyledons. These stores feed early growth before the leaves can make enough food.